

Patient Readmission Risk Initiative

Final Business Report | Steps 1-6 | 2 September 2026 | Target: 30-day readmission

Executive Decision Summary

The clinical data pipeline processed 1.155 million patient records and produced a standardized, model-ready dataset of 971,920 labeled patients. Data quality improvements reduced the feature space from 553 to 183 columns by normalizing Gender and grouping inconsistent medication names into 22 therapeutic classes. Three predictive models were executed successfully, but all produced AUC scores near 0.50, so the current data should support further investigation rather than operational deployment.

Business Scorecard

1.155M	971,920	15.0%	183
records analyzed	labeled clean records	30-day readmission rate	final model features

What Was Delivered

- **Step 1 - EDA:** identified missingness, invalid values, duplicate fields, two fully null columns, and class imbalance (84.98% not readmitted vs 15.02% readmitted). Generated distribution and correlation visuals.
- **Step 2 - Cleaning:** removed 37,988 invalid Age/Weight rows, 81,110 unlabeled rows, and 63,982 duplicates; median/mode imputation and IQR outlier capping were applied.
- **Data quality fix:** standardized Gender to Male/Female/Other/Unknown and corrected 50+ medication spellings into 22 therapeutic classes. Output: data_cleaned_v2.csv, 971,920 x 183.
- **Step 3 - Features:** created kidney stage, liver risk, polypharmacy, BMI category, age group, elderly high-dose, and AST/ALT ratio indicators. Output: data_engineered_v2.csv, 971,920 x 190.

Model Results: Readmission Prediction

Model	Accuracy	AUC-ROC	Readmitted Recall	Readmitted F1
Decision Tree	0.3007	0.5021	0.79	0.25
Random Forest	0.5699	0.5105	0.42	0.23
XGBoost	0.5317	0.5057	0.46	0.23

Business interpretation: AUC values are close to random-chance performance (0.50). The models are not ready for clinical decisions, financial forecasting, or automated intervention without improving the underlying signal and validating against an independent dataset.

Business Value and Recommended Actions

- **Value created:** a repeatable data foundation, explainable clinical risk indicators, cleaner medication taxonomy, and validated model artifacts for controlled experimentation.
- **Immediate action:** investigate target quality, feature/target relationships, temporal leakage, label definition, and whether the source data contains enough predictive signal.
- **Before deployment:** benchmark against a majority-class baseline, use temporal or external validation, review precision-recall tradeoffs, and obtain clinical governance approval.
- **Artifacts:** decision_tree_model.pkl, random_forest_model_v2.pkl, xgboost_best_model_v2.pkl, feature-importance plots, EDA visuals, and two-page-ready summary reports are available in outputs/.

Recommendation: proceed to model diagnostics and data-source improvement before production use.